

Superstore Sales Analytics & Prediction System

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Tools Used: Python | SQL | MySQL | Power BI | Machine Learning

GitHub Repository:

[\[View Project Repository\]](#)

1. Project Overview

Retail businesses generate large volumes of sales data, making it difficult to identify profitability trends, customer purchasing patterns, and regional performance manually. This project focuses on analyzing retail sales data using SQL, Python, and Power BI to generate business insights, visualize KPIs, and compare machine learning models for sales prediction.

2. Objectives

- Analyze sales and profit trends across categories and regions.
- Identify the impact of discounts on profitability.
- Perform SQL-based business analysis.
- Build predictive models for sales forecasting.
- Develop interactive Power BI dashboards for KPI reporting.

3. Dataset Information

- Dataset: Superstore Dataset
- Records: 9,000+
- Features: Sales, Profit, Category, Region, Discount, Quantity
- Target Variable: Sales

4. Tools & Technologies

- Python – Data Cleaning & Analysis
- SQL/MySQL – Business Queries & Database Analysis
- Power BI – Dashboard Development

- Pandas & NumPy – Data Manipulation
- Scikit-learn – Machine Learning
- Matplotlib & Seaborn – Data Visualization

5. Data Cleaning & Preprocessing

Performed data cleaning using Pandas to remove duplicates, handle missing values, and standardize dataset formatting. Converted categorical and numerical fields into analysis-ready formats for further processing.

6. Exploratory Data Analysis (EDA)

A. Sales by Category Analysis

Objective:

Analyze category-wise sales and profit performance to identify the most profitable and revenue-generating product categories.

Visualization:

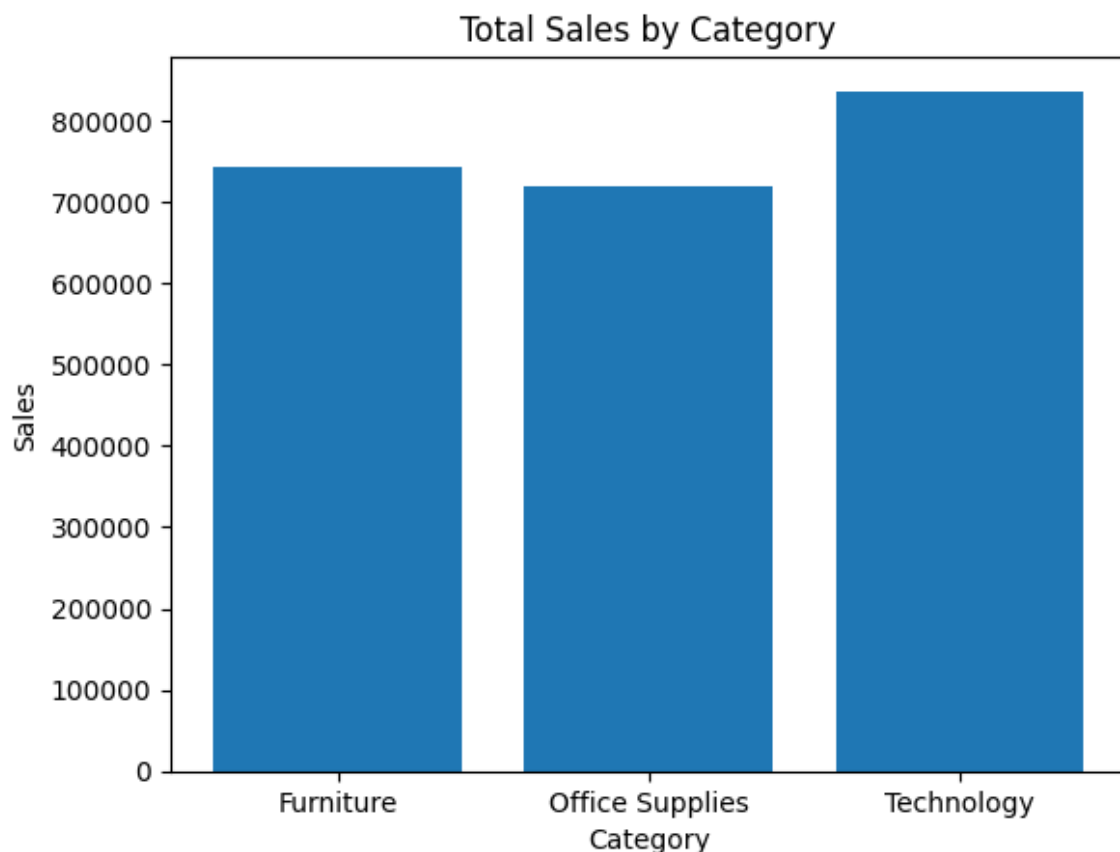


Figure 1: Sales by Category Analysis

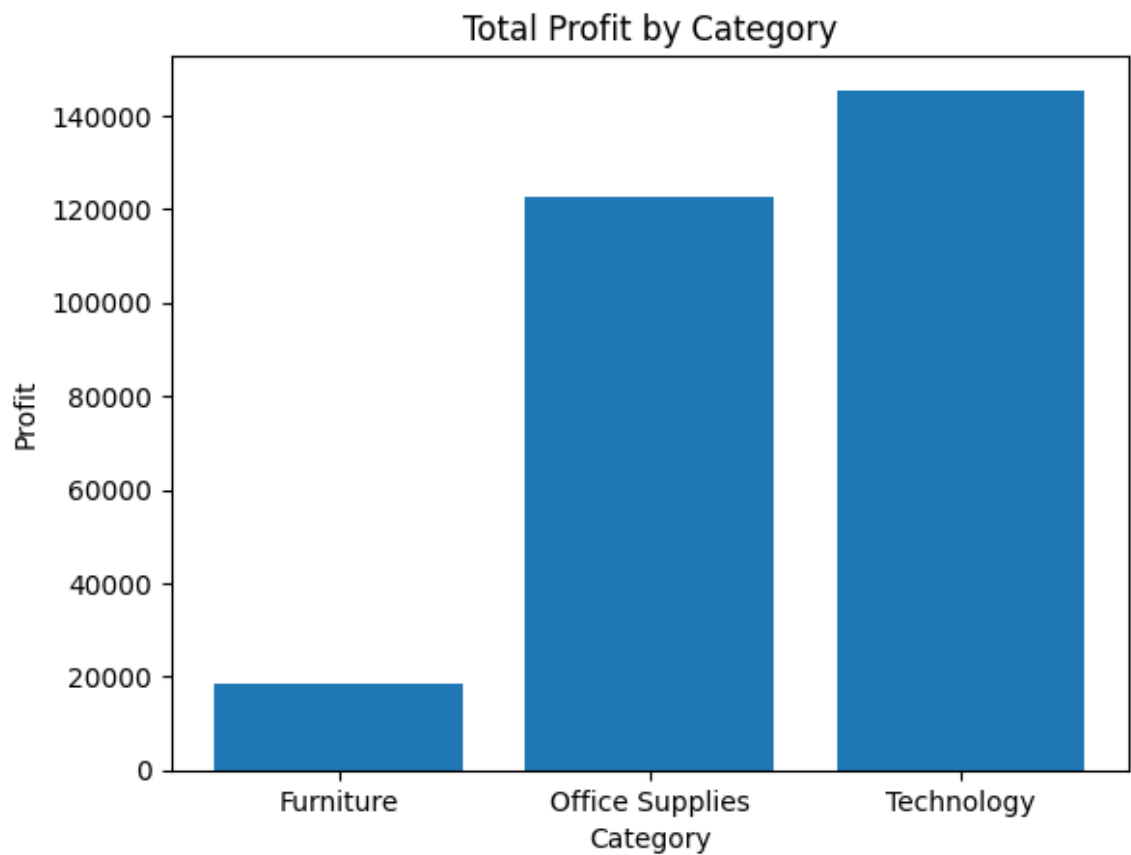


Figure 2: Profit by Category Analysis

Key Insights:

- Technology generated the highest sales and profit.
- Furniture showed high sales but lower profitability.
- Office Supplies maintained stable performance.

B. Profit by Region Analysis

Objective:

Evaluate regional sales and profit distribution to identify high-performing and low-performing business regions.

Visualization:

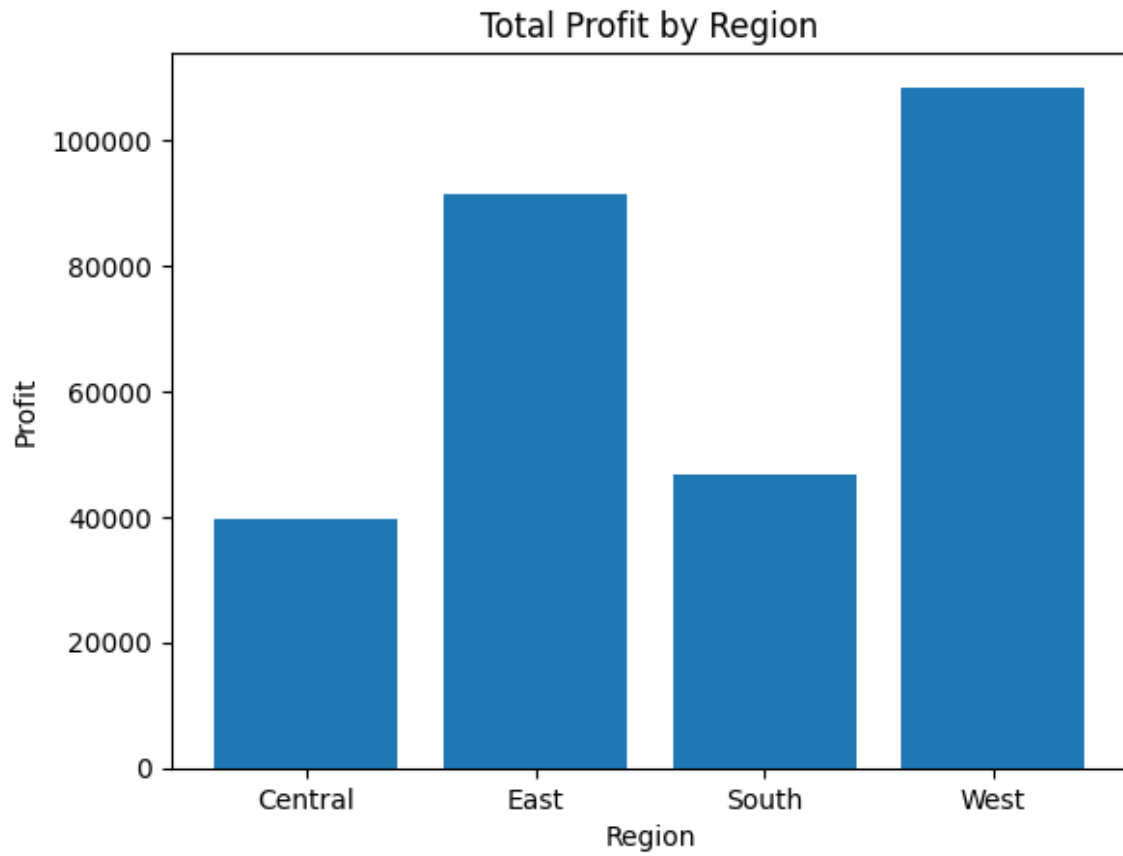


Figure 3: Profit by Region Analysis

Key Insights:

- Western region recorded the highest profit.
- Central region showed weaker profitability.
- Profit performance varied significantly across regions.

C. Discount vs Profit Analysis

Objective:

Understand the relationship between discount percentages and profitability to identify the impact of excessive discounting on business performance.

Visualization:

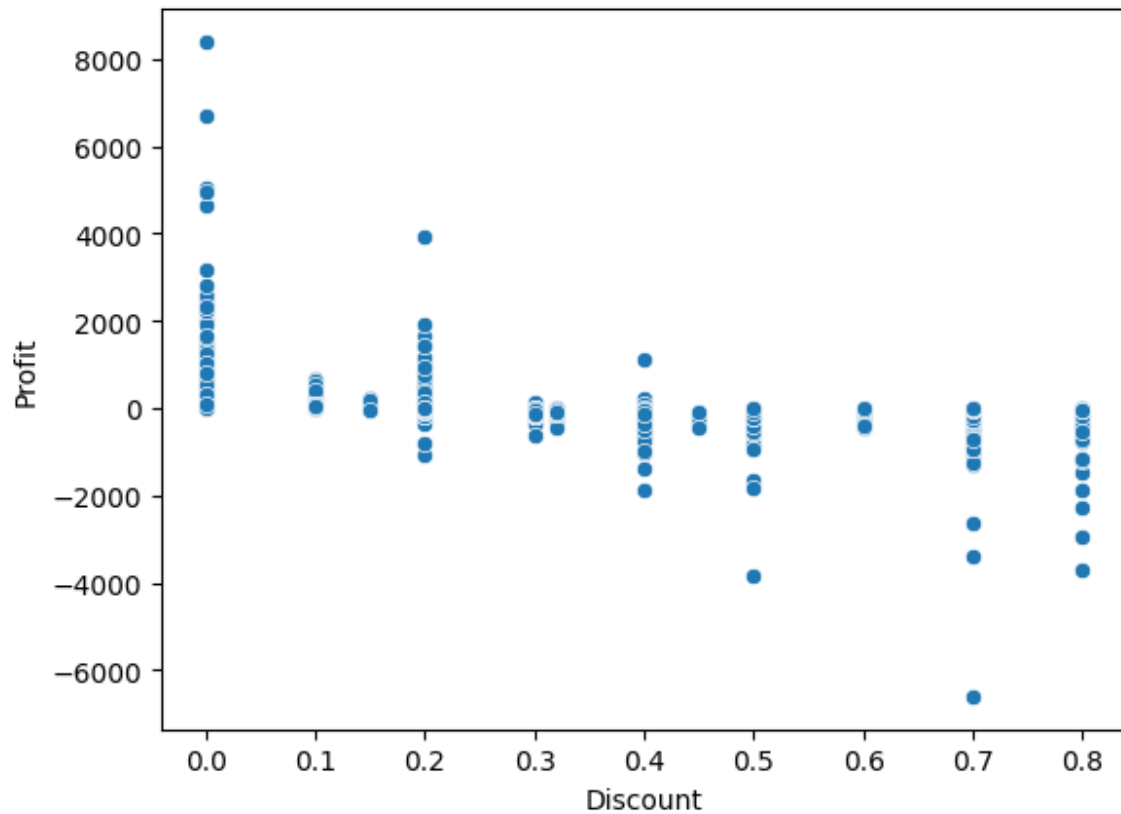


Figure 4: Discount vs Profit Analysis

Key Insights:

- Higher discounts reduced overall profitability.
- Several high-discount orders resulted in losses.
- Moderate discounts maintained better profit margins.

D. Correlation Heatmap

Objective:

Identify relationships between sales, profit, and discount variables to support business analysis and feature understanding.

Visualization:

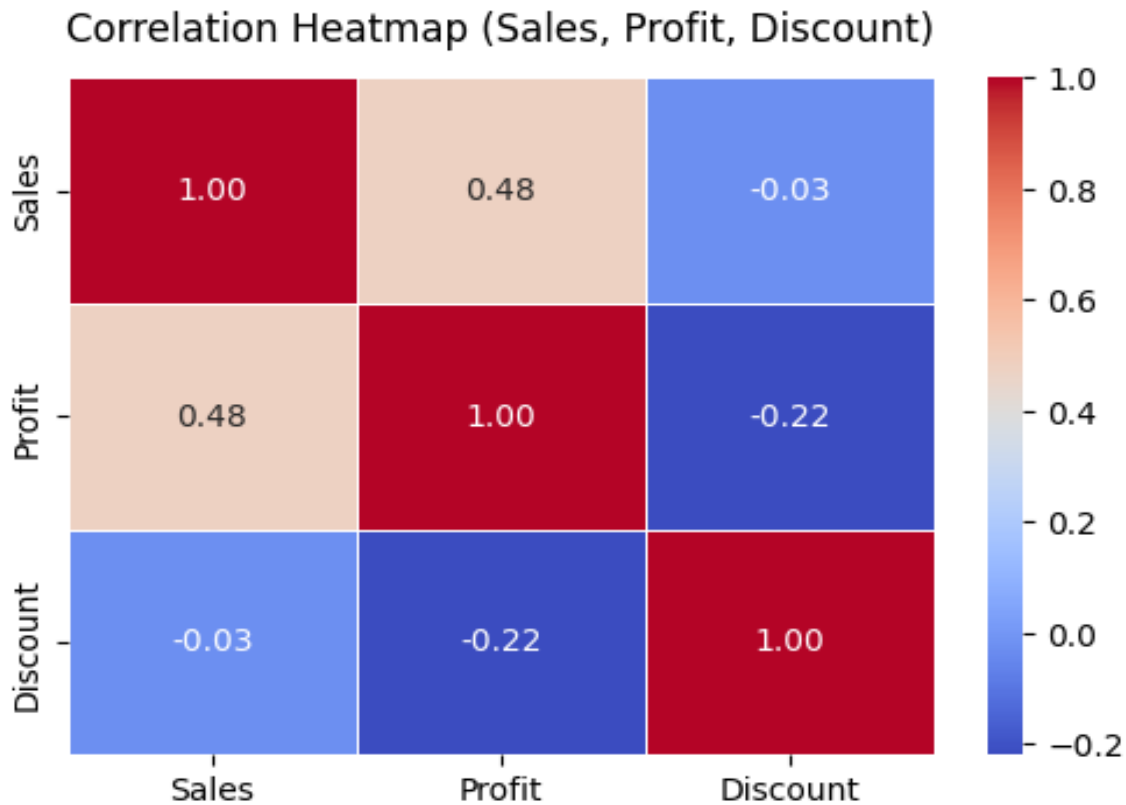


Figure 5: Correlation Heatmap

Key Insights:

- Sales and profit showed a positive correlation.
- Discount negatively impacted profit margins.
- Higher discount values were associated with lower profitability.

E. Monthly Sales Trend Analysis

Objective:

Analyze monthly sales patterns to identify seasonal trends, sales fluctuations, and business growth opportunities over time.

Visualization:

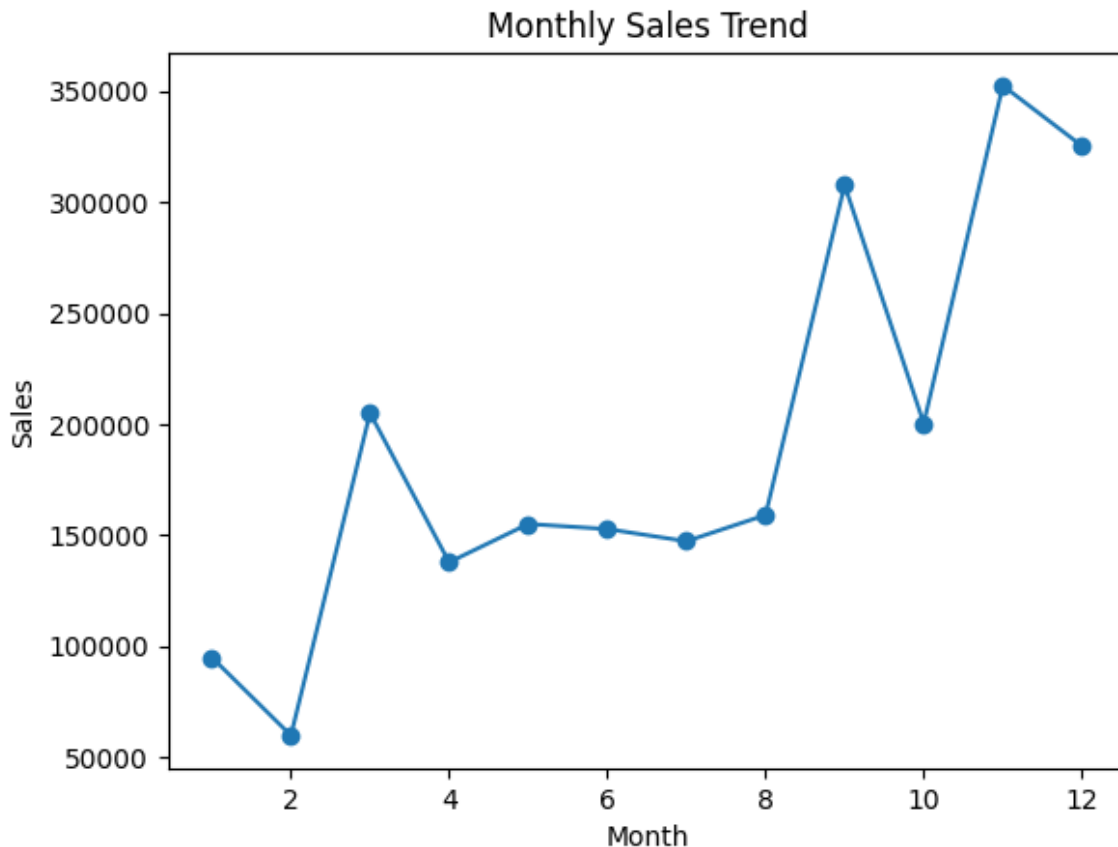


Figure 6: Monthly Sales Trend Analysis

Key Insights:

- Sales fluctuated across different months.
- Certain periods showed peak sales performance.
- Seasonal trends indicated opportunities for targeted marketing.

7. SQL Business Analysis

Developed analytical SQL queries using JOINS, aggregations, filtering, and grouping techniques to identify top-performing categories, regional sales trends, and profitability insights.

A. Monthly Sales Trend Analysis

Objective:

Analyze monthly sales patterns to identify seasonal trends and sales fluctuations over time.

SQL Query:

```
-- Monthly Sales Trend
select
    year(`Order Date`) as year,
    month(`Order Date`) as month,
    sum(Sales) as monthly_sales
from superstore_cleaned
group by year, month
order by year, month;
```

Output:

	year	month	monthly_sales
▶	2014	1	14188
	2014	2	4492
	2014	3	55602
	2014	4	28215
	2014	5	23578
	2014	6	34519
	2014	7	33859
	2014	8	27820
	2014	9	81615
	2014	10	31357
	2014	11	78434
	2014	12	69384
	2015	1	18145
	2015	2	11909

Figure 7: Monthly Sales Trend SQL Analysis

Key Insights:

- Sales performance fluctuated across different months.
- Certain periods showed significantly higher sales activity.
- Seasonal trends indicated opportunities for targeted marketing campaigns.

B. Regional Performance Analysis

Objective:

Evaluate regional sales performance and customer distribution across different business regions.

SQL Query:

```
-- Regional Performance (Group by + Having)
select
    `Region`,
    sum(Sales) as revenue,
    count(distinct `Customer ID`) as customers,
    avg(Sales) as avg_order
from superstore_cleaned
group by `Region`
having sum(Sales) > 100000
order by revenue desc;
```

Output :

	Region	revenue	customers	avg_order
►	West	723573	686	225.90477677177645
	East	677079	674	237.73841292134833
	Central	499887	629	215.19027120103314
	South	390765	512	241.21296296296296

Figure 8: Regional Performance SQL Analysis

Key Insights:

- Western region generated the highest overall revenue.
- Customer distribution varied significantly across regions.
- Average order values differed across business markets

C. Top 10 Sub-Categories by Revenue

Objective:

Identify the highest revenue-generating product sub-categories and evaluate their profit margins.

SQL Query:

```
-- Top 10 Sub-Categories by Revenue (Group by + Order by)
select
    `Sub-Category`,
    count(*) as orders,
    sum(Sales) as total_revenue,
    round(sum(Profit)/sum(Sales) * 100, 2) as profit_margin
from superstore_cleaned
group by `Sub-Category`
order by total_revenue desc
limit 10;
```

Output:

	Sub-Category	orders	total_revenue	profit_margin
►	Phones	889	329342	13.52
	Chairs	617	328097	8.1
	Storage	846	223368	9.53
	Tables	319	206798	-8.57
	Binders	1523	202593	14.92
	Machines	115	189155	1.79
	Accessories	775	166856	25.13
	Copiers	68	149462	37.21
	Bookcases	228	114728	-3.03
	Appliances	466	107266	16.91

Figure 9: Top Sub-Categories by Revenue

Key Insights:

- Certain sub-categories generated high revenue with stronger profit margins.
- High order volume did not always result in higher profitability.
- Profit margin analysis highlighted differences in category performance.

D. Top Customers Analysis

Objective:

Analyze customer purchasing behavior to identify high-value customers based on sales contribution.

SQL Query:

```
-- Top 10 Customers (Customer Segmentation)
select
    `Customer ID`,
    count(*) as order_count,
    sum(Sales) as lifetime_value,
    max(`Order Date`) as last_order
from superstore_cleaned
group by `Customer ID`
order by lifetime_value desc
limit 10;
```

Output:

	Customer ID	order_count	lifetime_value	last_order
▶	SM-20320	15	25035	2017-10-12
	TC-20980	12	19046	2016-11-26
	RB-19360	18	15107	2017-09-25
	TA-21385	10	14589	2017-10-22
	AB-10105	20	14460	2017-11-19
	KL-16645	29	14157	2017-11-13
	SC-20095	22	14127	2017-01-15
	HL-15040	11	12865	2017-11-17
	SE-20110	19	12198	2017-12-21
	CC-12370	11	12122	2017-11-17

Figure 10: Top Customers Analysis

Key Insights:

- A small group of customers contributed significantly to total sales revenue.
- High-value customers demonstrated repeated purchasing behavior.
- Customer-level analysis supported business segmentation insights.

E. Top Product per Category (Window Function)

Objective:

Identify the top-performing sub-category within each product category using SQL window functions.

SQL Query:

```
-- Top product per category (Window Functions)
with ranked as (
  select
    `Category`,
    `Sub-Category`,
    sum(Sales) as revenue,
    row_number() over (partition by `Category` order by sum(Sales) desc)
  as rk
  from superstore_cleaned
  group by `Category`, `Sub-Category`
)
select * from ranked where rk = 1;
```

Output:

	Category	Sub-Category	revenue	rk
►	Furniture	Chairs	328097	1
	Office Supplies	Storage	223368	1
	Technology	Phones	329342	1

Figure 11: Top Product per Category using Window Function

Key Insights:

- Top-performing sub-categories varied across product categories.
- Window functions improved analytical depth for category-level ranking analysis.
- Technology-related sub-categories generated strong revenue performance.

8. Machine Learning Analysis

Objective:

Build and compare regression models to analyze sales prediction performance using historical retail sales data.

Data Preparation

- Performed train-test split for model evaluation.
- Applied preprocessing and feature analysis on sales-related variables.
- Prepared analysis-ready datasets using Pandas and NumPy.

Models Used

The following regression models were implemented and evaluated:

- Linear Regression
- Random Forest Regressor
- Gradient Boosting Regressor
- XGBoost Regressor

Model Performance Comparison

```
Final Comparison:
Gradient Boosting -> R2: 0.217, RMSE: 679.98
Linear Regression -> R2: 0.184, RMSE: 694.30
Random Forest    -> R2: 0.230, RMSE: 674.64
XGBoost          -> R2: 0.142, RMSE: 711.96
```

Visualization:

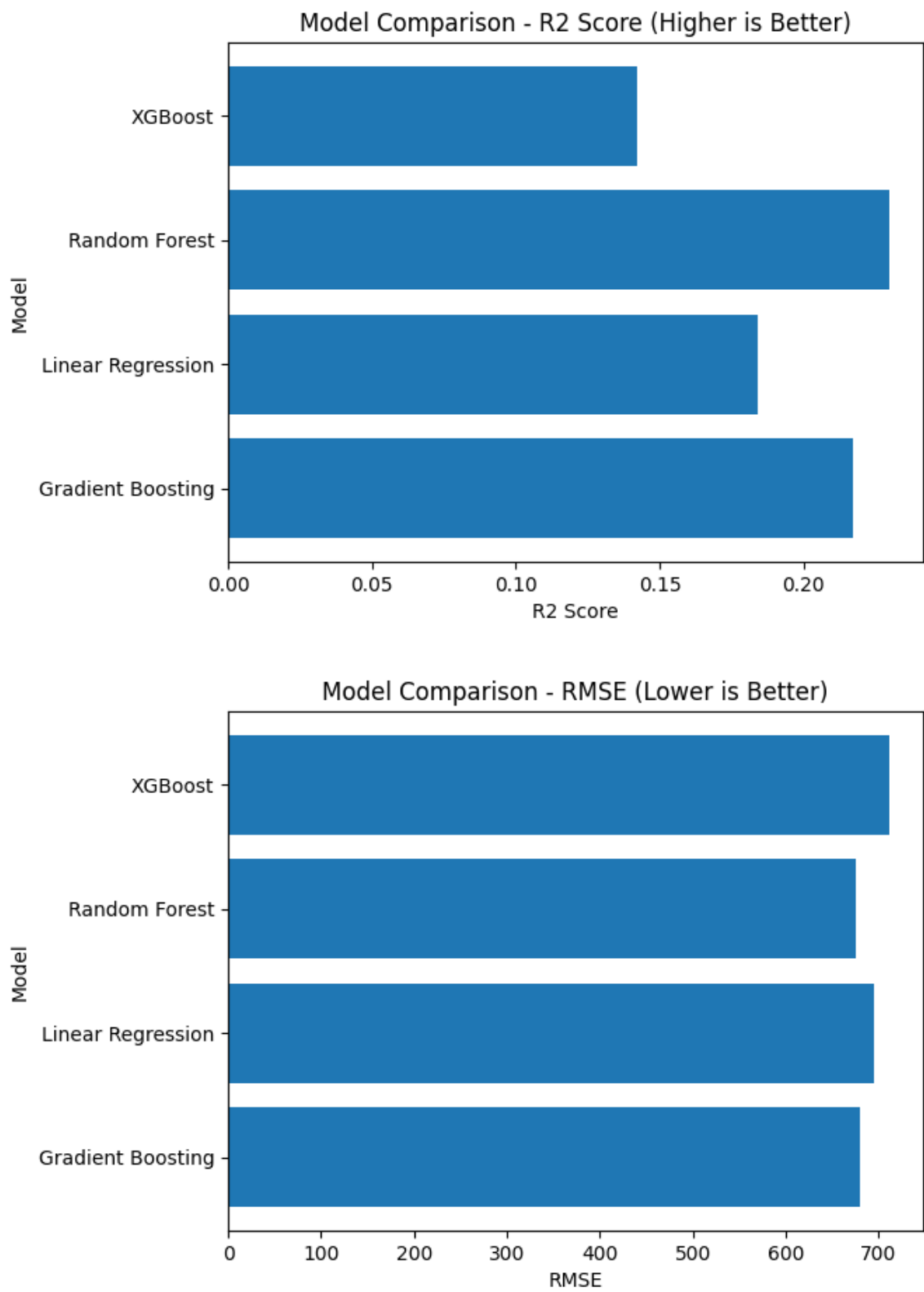


Figure 12: Machine Learning Model Performance Comparison

Key Insights:

- Random Forest achieved the best predictive performance among the evaluated models.
- Ensemble models performed better than Linear Regression on retail sales data.
- Predictive accuracy remained moderate due to dataset variability and limited external business features.

9. Power BI Dashboard

Objective:

Develop an interactive business intelligence dashboard to monitor sales performance, profitability, regional trends, and category-wise business insights.

Dashboard Overview

The dashboard was designed using Power BI to transform raw retail sales data into interactive business insights through KPI tracking, visual analysis, and dynamic filtering.

Dashboard Features

- KPI cards for Total Sales, Total Profit, and Total Orders
- Category-wise sales and profit analysis
- Regional sales performance visualization
- Monthly sales trend monitoring
- Interactive slicers and filters for dynamic reporting

Dashboard Visualization

Superstore Sales Analysis Dashboard

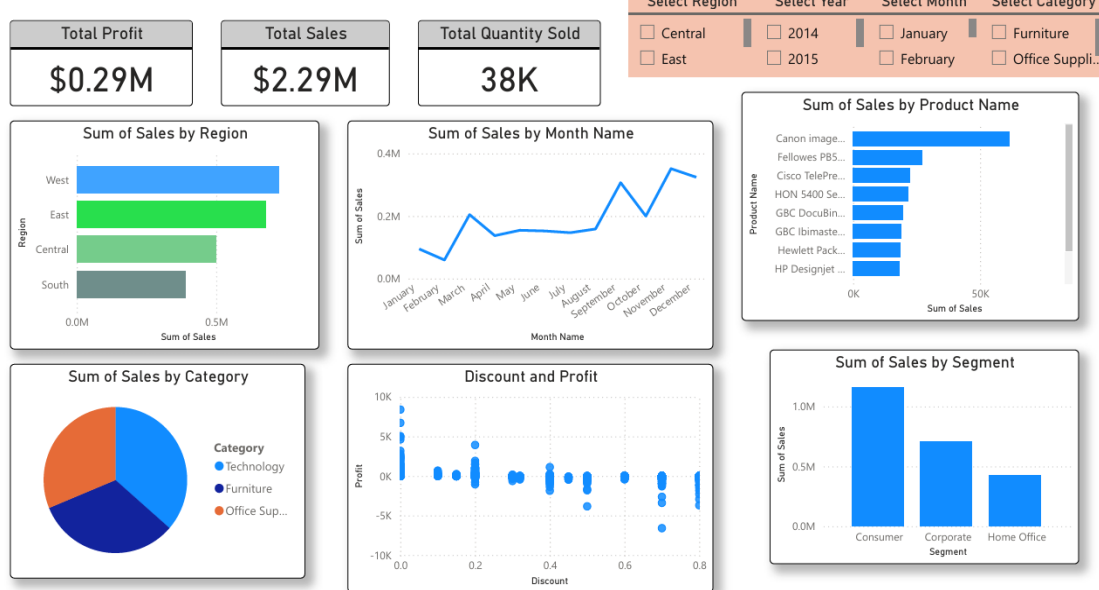


Figure 13: Power BI Sales Dashboard

Key Insights:

- Technology category generated the highest sales performance.
- Western region contributed the highest overall profit.
- High discounts negatively impacted profitability.
- Sales performance fluctuated across different months.
- Interactive filtering improved business analysis across regions and categories.

Business Impact:

- Improved business reporting through interactive visualization and KPI monitoring.
- Enabled faster analysis of sales trends and regional performance.
- Supported data-driven decision-making through dynamic dashboard insights.

10. Key Business Insights

- Business profit depended not only on sales but also on discount management and product performance.
- Sales and profit varied across different regions and product categories.
- Customer analysis helped identify high-value customers contributing significantly to revenue.
- Interactive dashboards improved visibility into sales trends, KPIs, and overall business performance.

11. Challenges & Limitations

- Retail sales are influenced by external business factors not available in the dataset.
- Limited feature engineering opportunities affected predictive model performance.
- Some regions and categories showed inconsistent profitability trends.
- The project focused on analytical workflow and business insights rather than production-level forecasting accuracy.

12. Conclusion

This project demonstrated an end-to-end data analytics workflow using SQL, Python, machine learning, and Power BI on a real-world retail sales dataset. The analysis helped identify sales trends, profitability patterns, regional performance, and the impact of discounts on business outcomes. Through data cleaning, exploratory analysis, SQL-based business reporting, predictive modeling, and dashboard development, the project strengthened practical understanding of data-driven decision-making and business analytics.